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Interface circuits for a six-phase variable reluctance energy harvester: a comparative study of rectification and power conditioning topologies

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*To my parents
and friends*

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Abstract

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Sommario

Negli ultimi anni la crescente diffusione di reti di sensori wireless (WSN) e di dispositivi IoT ha reso critico il problema della loro alimentazione: le soluzioni a batteria comportano infatti costi di sostituzione e manutenzione non trascurabili, specialmente quando i nodi sensore sono numerosi o di difficile accesso. L'*energy harvesting*, ossia la conversione di energia ambientale – vibrazioni meccaniche, calore, luce, segnali RF – in energia elettrica utilizzabile, si propone come soluzione a questo problema, permettendo la realizzazione di nodi sensore autonomi e a manutenzione pressoché nulla.

Questo lavoro di tesi si apre con una panoramica delle principali tecnologie di trasduzione per l'*harvesting* da vibrazioni meccaniche – piezoelettrica, elettromagnetica, elettrostatica e triboelettrica – evidenziandone vantaggi e limiti, per poi concentrarsi sui raccoglitori a riluttanza variabile (*Variable Reluctance Energy Harvester*, VREH). Questa tecnologia è particolarmente adatta alle applicazioni rotazionali a bassa velocità, poiché magnete e bobina restano entrambi fissi e la variazione di flusso concatenato è generata unicamente dal passaggio di una ruota dentata ferromagnetica. Viene quindi ripercorso lo stato dell'arte della ricerca su questa tecnologia, dai primi prototipi per il monitoraggio ferroviario fino alle topologie a polo "a m" ottimizzate per la densità di potenza volumetrica, sino al raccoglitore multifase (MP-VREH) a sei unità di pickup preso come riferimento in questo lavoro, pensato per l'integrazione in un cuscinetto di ruota di veicoli commerciali di grandi dimensioni, quali autobus, a velocità di rotazione comprese tra 100 e 400 giri/min (circa 20–80 km/h).

A partire dalla legge di Faraday viene poi sviluppato un modello elettrico completo del VREH, con la derivazione delle relazioni che legano la potenza estratta ai parametri geometrici ed elettrici del trasduttore – numero di spire, resistenza di bobina, differenza di flusso tra posizione allineata e disallineata, numero di denti della ruota – fino alla determinazione della potenza massima teoricamente trasferibile al carico e alla discussione delle condizioni di adattamento di impedenza in funzione della velocità di rotazione.

La parte centrale della tesi è dedicata alla progettazione dei circuiti di condizionamento della potenza per il raccoglitore a sei fasi considerato. Vengono proposte e confrontate diverse topologie di raddrizzamento, valutate dapprima tramite simulazione circuitale e successivamente tramite la realizzazione di un banco di prova sperimentale, comprensivo della progettazione di una scheda elettronica dedicata e della taratura di un sensore di corrente per la misura accurata della potenza estratta dal sistema. Infine, viene affrontata la scelta del circuito integrato di power management (PMIC) più adatto a regolare e immagazzinare l'energia raccolta, completando così la catena di conversione dall'energia meccanica rotazionale fino all'alimentazione stabile del carico elettronico finale.

Introduction

1.1 Motivation

The number of wireless sensor networks (WSNs) and Internet of Things (IoT) devices has grown rapidly in recent years, raising a fundamental question: how to power all these devices without replacing batteries or requiring constant maintenance. Recent advances in low-power electronics have reduced the energy consumption of sensor nodes to the range of tens to hundreds of microwatts [10]. At such low power levels, it becomes possible to power these devices by extracting energy from the surrounding environment, removing the need for batteries and allowing the network to operate indefinitely.

Batteries are the most common power source for WSN nodes, but they have well-known limitations. In many applications, such as structural health monitoring, industrial machinery surveillance, or medical implants, replacing or recharging depleted batteries is difficult or simply not possible [5]. Using larger batteries to extend the device lifetime increases the size of the system, which is often not acceptable. In addition, the environmental cost of battery and the consequent maintenance of large-scale networks are serious obstacles to the widespread use of WSNs [8].

Energy harvesting offers a practical solution to these problems. Instead of storing a fixed amount of energy, harvesting systems collect energy from sources available in the environment such as sunlight, heat gradients, mechanical vibrations, or radio frequency signals and convert it into electrical power to supply an electronic system.

Among all available energy sources, mechanical vibrations have received the most attention in the research community, accounting for more than half of

publications in this field [8]. Piezoelectric, triboelectric, and electromagnetic transducers are the most widely used mechanisms, each with different trade-offs in output voltage, power density, and complexity [10, 5]. Figure 1.1 shows the distribution of recent publications about transducer types, highlighting the dominance of mechanical energy harvesting approaches.

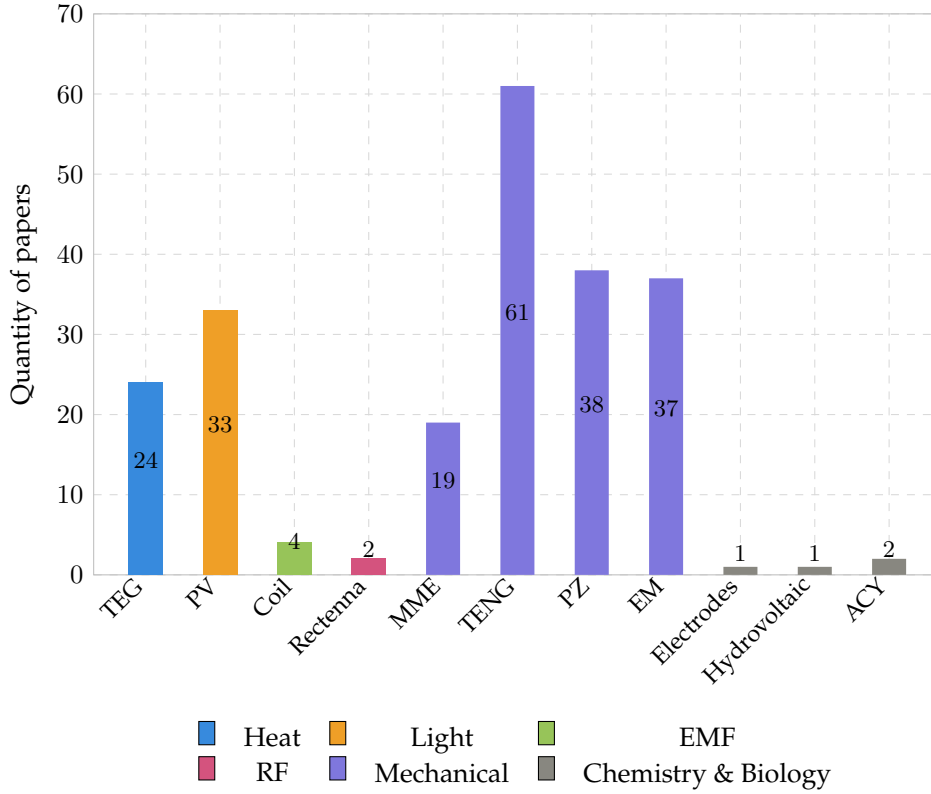


Figure 1.1: Distribution of papers based on transducer type [8].

Energy harvesting systems are generally divided into two categories [5]: systems that use the harvested energy immediately (*Harvest-Use*), and systems that store it first in a battery or supercapacitor before supplying it to the node (*Harvest-Store-Use*). The second approach is far more common, since most energy sources are not continuously available and some form of storage is needed to guarantee uninterrupted operation.

1.2 Energy Harvesting Systems

An energy harvesting system is made of three components that work in sequence, as illustrated in Figure 1.2.

- *Transducer*: converts the ambient energy into an electrical signal.
- *Power conditioning circuit*: perform AC-DC conversion, impedance matching and voltage regulation. A typical implementation combines a rectifier circuit with a power management integrated circuit (PMIC) [14].
- *energy storage element*: is used to keep the device operating for a certain amount of time even if the source is weak or no more present. Such element can be a capacitor/super capacitor or a rechargeable battery [10].

Figure 1.2: Block diagram of a typical energy harvesting system.

The performance of the entire system depends in particular on the transducer. Its design must be matched to the characteristics of the ambient source as well as to the power need of the target device. A transducer that is well suited to one environment may perform poorly in another, which is why the choice of transduction mechanism is the central design decision in any harvesting application [5].

Designing a good harvesting system is not just about getting the most power out of it. The system must work reliably in real conditions, where the energy source can be irregular or unpredictable. At the same time, it must be small, cheap, and easy to integrate into the target device [5]. Finding the right balance between these requirements is the main challenge in this field, and it has led to the development of many different solutions, as discussed in the following sections.

1.3 Performance metrics

A key challenge in evaluating energy harvesting systems is choosing the right performance metrics. Xu et al.[14] propose two metrics for this purpose. The first is the *power conversion efficiency* (PCE), which measures how much of the power entering the interface circuit actually reaches the output:

$$\text{PCE} = \eta_{\text{Rec}} \cdot \eta_{\text{DC-DC}} = \frac{P_{\text{out}}}{P_{\text{in}}}. \quad (1.1)$$

The second is the *power extraction efficiency* (PEE), which compares the output power to the maximum power the transducer could deliver under ideal conditions:

$$\text{PEE} = \frac{P_{\text{out}}}{P_{\text{max}}} = \frac{P_{\text{in}}}{P_{\text{max}}} \cdot \text{PCE}. \quad (1.2)$$

The difference between the two is important. A circuit can convert power efficiently (high PCE) while still wasting much of what the transducer could produce, if the circuit itself is not able to work in the optimal operating point. For this reason, PEE is a more useful metric when the goal is to maximise the total power delivered to the load [14].

1.4 Main Transduction Technologies

Mechanical vibrations can be converted into electrical energy through four main transduction mechanisms: piezoelectric, electromagnetic, electrostatic, and triboelectric. Each of them works in a different way and comes with its own advantages and limitations [9, 2].

Piezoelectric

Piezoelectric transducers produce a voltage when a material is mechanically deformed. The most common design is a cantilever beam with a piezoelectric layer on top: when the beam vibrates, the material deforms and generates an electric charge. This approach gives a high output voltage and does not need a magnet or moving parts, which makes it easy to integrate into small devices [1]. The main drawback is that piezoelectric materials are fragile and may crack under large deformations. Also, the output drops quickly when the vibration frequency moves away from the resonant frequency of the beam [10].

Electromagnetic

Electromagnetic transducers generate a voltage by moving a coil near a permanent magnet. This is the same principle used in large electrical generators, applied at a small scale. These harvesters tend to produce a low voltage but can deliver a relatively high current, which makes them a good choice when power output is the main goal. One limitation is that the coil and magnet must move relative to each other, which can be hard to achieve in applications involving large or slowly rotating objects [9].

Electrostatic

Electrostatic transducers work by changing the capacitance between two conductive plates as they move. When the plates get closer or further apart due to vibration, the stored electrical energy changes and can be extracted as useful power. This approach is easy to manufacture using standard microfabrication processes, making it attractive for very small devices. However, it requires an

external voltage or a pre-charged material to get started, which adds complexity to the system [9, 2].

Triboelectric

Triboelectric transducers use contact between two different materials to generate surface charges. When the materials touch and then separate, charges build up and can drive a current through an external circuit. This type of harvester, known as a triboelectric nanogenerator (TENG), was first demonstrated by Fan et al.[4] and has attracted a lot of interest because it can be made from cheap, flexible materials and works well at low frequencies, such as those produced by human motion. The main weaknesses are sensitivity to humidity and wear of the contact surfaces over time [8].

Among these four mechanisms, no single technology is universally superior. The best choice depends on the specific application, the characteristics of the available vibration source, and the constraints on size and cost. This comparison motivates the investigation of alternative approaches, including variable reluctance transduction, which is discussed in the following section.

1.5 The Variable Reluctance Principle

Variable reluctance energy harvesting (VREH) is a type of electromagnetic energy harvesting based on Faraday's law of induction. The transducer consists of three main parts: a permanent magnet, a pickup coil wound around a ferromagnetic pole piece, and a toothed ferromagnetic wheel that rotates nearby. When a tooth on the wheel aligns with the pole piece, the magnetic flux through the coil reaches its maximum value. When the gap between two teeth faces the pole piece, the flux drops to its minimum. This periodic variation in flux induces an alternating voltage across the coil, with a frequency that depends on the rotational speed and the number of teeth on the wheel [12]:

$$f_c = \omega N_t, \quad (1.3)$$

where ω is the rotational speed in rad/s and N_t is the number of teeth on the wheel. Following Faraday's law, the output power of the transducer delivered to a matched load can be expressed as [13]:

$$P_L = \frac{N_{\text{Coil}}^2}{8R_{\text{Coil}}} \Delta_\phi^2 \left[\frac{\omega N_{\text{Tooth}}}{2\pi} \right]^2, \quad (1.4)$$

where N_{Coil} is the number of coil turns, R_{Coil} is the coil resistance, $\Delta\phi$ is the magnetic flux difference between the aligned and unaligned positions, and N_{Tooth} is the number of teeth. This expression shows that the output power grows with the square of the rotational speed, and that it can be increased by maximising the flux difference and the number of coil turns, or by reducing the coil resistance.

What makes VREH different from most other electromagnetic harvesters is that neither the magnet nor the coil needs to move. The flux variation is caused entirely by the relative motion between the toothed wheel and the pickup units. This removes the need for a mechanical connection between the rotating part and the harvester, which is an advantage. It also means that the device is easy to scale: changing the shaft diameter mainly affects the size of the toothed wheel, while the pickup unit can stay the same [13]. In contrast, traditional electromagnetic generators require scaling of the magnets or coils, which becomes costly for large shafts or slow rotational speeds.

At low rotational speeds, the output voltage and frequency are both small, which makes the design of the interface circuit more critical [14]. At the same time, low-speed applications are exactly where VREH has the clearest advantage over other electromagnetic harvesters, since these typically require a relative displacement between coil and magnet that is hard to achieve at low speeds [9].

Its application to energy harvesting is more recent. Kroener et al. [7] were among the first to study a VREH specifically designed for energy harvesting in railway monitoring, showing that the principle could reliably power wireless sensor nodes from the motion of passing train wheels. Since then, the approach has been explored in a growing number of industrial and structural monitoring applications [12, 13].

1.6 State of the Art of VREH

The first applications of the variable reluctance principle to energy harvesting appeared in the context of railway monitoring. Kroener et al. [7] demonstrated that a VREH placed beside a rail track could generate enough power from the passage of train wheels to supply a wireless sensor node, reporting a peak output power of around 5.9 mW at a simulated train speed of 81.5 Km/h. This work showed for the first time that the variable reluctance principle could be practically used as an energy source rather than just a measurement transducer.

A systematic comparison of VREH topologies was later carried out by Xu et al. [12], who surveyed the structures proposed in the literature and evaluated them in terms of output power and power density for low-speed rotating applications. The study identified the *m-shaped* pickup unit, which uses two opposing

permanent magnets and a centre pole piece, as the topology with the highest volumetric power density among all structures considered. This result motivated a detailed design study of the m-shaped VREH, in which the influence of three key geometric parameters was investigated: the coil height h_{Coil} , the tooth height h_{Tooth} , and the magnet height h_{M} [13]. The study showed that these parameters involve trade-offs under size constraints, and provided a numerical optimisation method based on finite element analysis (FEA). The optimised structure reached output powers between 0.12 mW and 22 mW at rotational speeds of 5 rpm to 60 rpm within a volume constraint of 43.5 cm³, which is sufficient to power a range of duty-cycled wireless sensor systems [13].

A practical challenge in VREH systems is that the transducer produces a low-voltage alternating output, which must be rectified and conditioned before it can supply any systems. At low rotational speeds, the output voltage can be well below 1 V, which makes the choice of rectifier circuit critical. Xu et al. [14] compared three passive rectifier topologies: a full-wave bridge rectifier (FWR), a voltage doubler (VD), and a negative voltage converter (NVC), all on the same VREH system at speeds between 5 rpm and 20 rpm. The results showed that the VD achieved the best power extraction efficiency at low speeds, reaching a PEE of 40-60%, while the NVC outperformed it above 18 rpm. The FWR showed the lowest performance across all conditions, with PEE values of only 20-40%. These results confirm that the interface circuit is not just a passive element but actively affects how much power the transducer can deliver to the load.

These studies show that VREH is a mature and well-understood technology for low-speed rotational applications, with demonstrated power levels sufficient for a variety of wireless sensing applications. The most recent step forward in this direction is the multi-phase VREH proposed by Wu et al. [11], which integrates six m-shaped pickup units into a bearing hub unit for large commercial vehicles, reaching output powers between 0.46 W and 4.77 W at rotational speeds of 100 to 400 RPM. This significant increase in output power opens new possibilities for powering more demanding condition monitoring systems. However, it also introduces new challenges in the design of the power conditioning circuit, which must handle six independent AC outputs with different phases and varying voltage levels depending on the rotational speed. These challenges are discussed in the next section.

1.7 Open Challenges

Despite the progress described in the previous sections, several challenges remain open in the design of VREH systems. This section focuses on those that

are most relevant to the work presented in this thesis.

Power conditioning for multi-phase outputs

Single-phase VREH systems produce one alternating voltage signal that must be rectified and converted into a stable DC supply. This problem has been studied in detail for low-speed single-phase harvesters, where the choice of rectifier topology has a significant impact on the power extraction efficiency [14].

Many studies have investigated the design of power conditioning circuits for single-phase energy harvesters. The most common approach combines a rectification stage with a DC-DC converter, such as a Buck or Boost circuit, to regulate the output voltage to a level suitable for the target load [15, 16, 6, 3]. However, these solutions are designed for a single AC input and cannot be directly applied to a multi-phase harvester such as the MP-VREH, where six independent AC signals must be processed simultaneously.

Furthermore, many of these solutions do not include a startup circuit or a control system, which are both essential for the reliable operation of the power conditioning circuit in practice. Without a startup circuit, the system is not able to start operating when the harvested voltage is initially too low to power the control electronics. Without a control system, the circuit cannot adapt to changes in the rotational speed or load conditions, which are essential in real-world applications.

This gap in the literature motivates the investigation presented in this thesis.

When the number of pickup units increases, as in the multi-phase VREH proposed by Wu et al. [11], which uses six m-shaped pickup units arranged around a common toothed wheel, the power conditioning problem becomes considerably more complex. Each pickup unit produces an independent AC output with a different phase and the six signals are evenly spaced in time by 60° phase shift. This means that standard single-phase rectifier solutions cannot be directly applied, and that the interface circuit must be designed to handle multiple AC inputs efficiently.

Low output voltage at low speed

A fundamental limitation of VREH transducers is that the output voltage scales with rotational speed. At low speeds, the induced voltage may be well below 1 V, which makes rectification difficult because conventional diode bridges lose a significant fraction of the available voltage across their forward drops [14]. In a multi-phase system, this problem affects all six phases simultaneously. Finding a rectifier topology that works efficiently across a wide speed range, from the

minimum speed at which the harvester can sustain the load, up to the maximum operating speed, is therefore a key design requirement.

Impedance matching under variable conditions

The maximum power that a VREH can deliver to a load is achieved when the load impedance is matched to the internal impedance of the pickup coil. This matching condition depends on the rotational speed, because the electrical frequency of the output signal changes with speed. For a single-phase harvester, Xu et al. [13] showed that a fixed resistive load leads to an approximately linear increase in output power with speed, whereas a complex conjugate matched load, which compensates for the inductive component of the coil, could further increase the output power, especially at higher speeds. In a six-phase system, achieving dynamic impedance matching for all phases simultaneously adds a further layer of complexity to the interface circuit design.

1.8 Harvester Specifications and Research Objective

The work presented in this thesis is based directly on the MP-VREH proposed by Wu et al. [11]. The electrical characteristics of a single pickup unit, obtained from the transducer model, are summarised in Table 1.1. These values are used throughout this work as the reference operating conditions for the design and evaluation of the proposed power conditioning circuits.

RPM	f (Hz)	V_{Lpk} (V)	R_{coil} (Ω)	L_{coil} (mH)	P_o (W)
100	36.7	0.91	6	21.68	0.41
200	73.3	1.35	6	21.46	0.91
300	110.0	1.59	6	21.27	1.26
400	146.7	1.74	6	21.12	1.51

Table 1.1: Electrical parameters of a single MP-VREH pickup unit at different rotational speeds.

All values in Table 1.1 were obtained under impedance-matched load conditions. V_{Lpk} refers to the peak voltage across the load, and P_o is the average power dissipated in the load.

These challenges motivate the work presented in this thesis, which focuses on the design and evaluation of power conditioning circuits for the six-phase MP-VREH described above. The goal is to identify the interface circuit topology that maximises the power delivered to the load across the relevant operating

speed range of 100 to 400 RPM (20-80 Km/h), corresponding to typical driving conditions of large commercial vehicles, for exaple buses.

Theoretical Background

This chapter introduces the physical and circuit-level concepts required to motivate the design choices made in the following chapters. In this chapter, starting from Faraday's law, an electrical model of the variable-reluctance energy harvester (VREH) is derived. The model is then used to discuss the maximum power transfer theorem and the role of impedance matching.

2.1 Electrical Model of the Variable-Reluctance Harvester

2.1.a Faraday's law

The Faraday's law is the starting point for any electromagnetic transducer, which relates the electromotive force $\varepsilon(t)$ (EMF) induced in a closed loop wire to the time rate of change of the magnetic flux $\Phi(t)$ linked to the loop.

$$\varepsilon(t) = -\frac{d\Phi(t)}{dt} \quad (2.1)$$

For a coil made by N_t turns sharing the same flux, the total linked flux is $\Lambda(t) = N \Phi(t)$ and the induced EMF becomes:

$$\varepsilon(t) = -\frac{d\Lambda(t)}{dt} = -N_t \frac{d\Phi(t)}{dt}. \quad (2.2)$$

In case of a coil consisting of N_t turns mechanically spaced one respect to each other leading to different linked flux, the induce voltage can be described by

$$\varepsilon(t) = -\frac{d}{dt} \cdot \sum_{n=i}^{N_t} \Phi_i(t) \quad (2.3)$$

where Φ_i is the flux linkage associated to a single turn of the coil [12].

2.1.b Variable Reluctance Principle

Unlike conventional electromagnetic transducers, in which the EMF is induced by a relative motion between a permanent magnet (PM) and a pickup coil, a variable-reluctance energy harvester (VREH) keeps both the magnet and the coil stationary. The magnetic flux linked to the coil is instead modulated by the passage of a rotating, toothed ferromagnetic wheel in front of a stationary pickup unit: as a tooth or a slot alternately faces the pickup unit, the reluctance of the magnetic circuit changes periodically, and it is this reluctance variation – rather than any bulk motion of the magnet – that drives the flux change $\Phi_i(t)$ in (2.3) [12]. Because the rotating part carries no magnet, no coil and no electrical connection, the resulting transducer is mechanically simple, robust, and easily rescaled to shafts of different diameters [13].

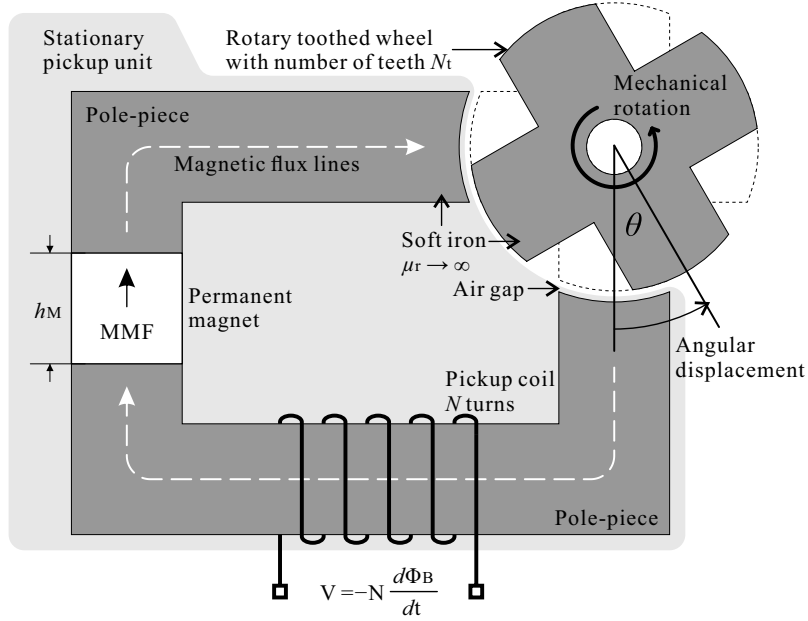


Figure 2.1: Structure of a variable-reluctance sensor: a stationary pickup unit, composed of a permanent magnet, a soft-ferromagnetic pole-piece and a pickup coil, faces a rotating toothed wheel across an air gap.

The pole-piece and the toothed wheel are both made of soft ferromagnetic material, while the PM is a hard ferromagnetic material providing a magnetomotive force (MMF) $\mathcal{F}_M$ that can be taken as constant. As the wheel rotates, the pickup unit cycles between two extreme relative positions: the *aligned* position,

where a pole of the pickup unit faces a tooth of the wheel and the air-gap reluctance is minimum, and the *unaligned* position, where the pole faces a slot and the reluctance is maximum [12, 13]. The corresponding flux swings between a maximum Φ_{max} and a minimum Φ_{min} , so that the relevant design quantity is the flux difference $\Delta\Phi = \Phi_{max} - \Phi_{min}$ between the two extremes. Fig. 2.2 illustrates this swing directly: the air-gap flux density is sharply peaked at the aligned position and nearly collapses at the unaligned position, confirming that the flux change is concentrated within a narrow angular window around each tooth-slot transition.

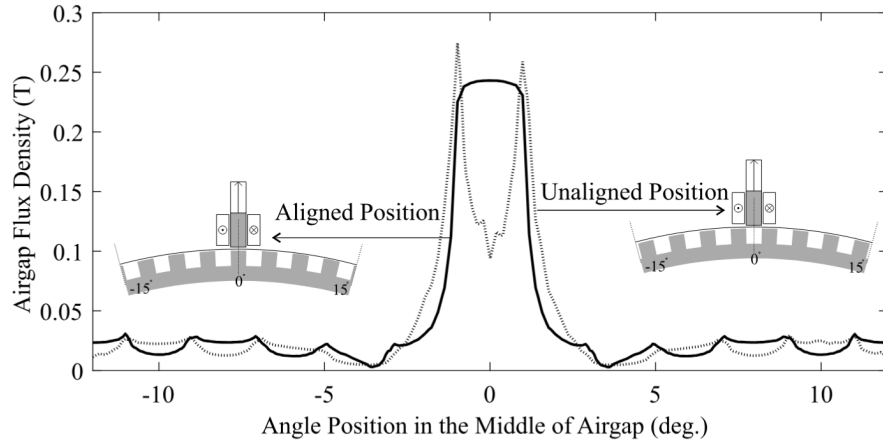


Figure 2.2: Airgap flux density distribution for aligned and unaligned positions. Reproduced from [12].

From flux swing to output power

For a wheel with N_t teeth rotating at angular velocity ω , the electrical frequency of the induced voltage is

$$f_c = \omega N_t. \quad (2.4)$$

At low rotational speeds the coil impedance is dominated by its resistance R_{Coil} , so that maximum power is transferred to a purely resistive load matched to it, $R_L = R_{Coil}$; under this condition the load voltage is related to the open-circuit coil voltage $V_c(t)$ of (2.3) by

$$V_L(t) = \frac{1}{2} V_c(t). \quad (2.5)$$

Assuming the time-varying flux to be an ideal sinusoidal wave, uniformly distributed in the N_c turns of the coil, the peak-to-peak voltage developed across

the matched load follows from (2.3) and (2.5) as [12]

$$V_{L,pp} = -\frac{1}{2} N_c \frac{\Phi_{max} - \Phi_{min}}{\frac{1}{2}T_c} = -N_c f_c \Delta\Phi, \quad (2.6)$$

where $\frac{1}{2}T_c$ is the time taken by the wheel to rotate from the aligned to the unaligned position, i.e. from a tooth to a slot centred on the pole-piece.

Because $V_{L,pp}$ in (2.6) is the peak-to-peak amplitude of an (assumed) sinusoidal load voltage, its root-mean-square value is $V_{L,rms} = V_{L,pp}/(2\sqrt{2})$, so that the RMS power dissipated in the matched load is [12]

$$P_L = \frac{V_{L,rms}^2}{R_L} = \frac{V_{L,pp}^2}{8 R_{Coil}} = \frac{(\omega N_t N_c)^2}{8 R_{Coil}} (\Delta\Phi)^2, \quad (2.7)$$

where the last equality follows from substituting (2.4) into (2.6). Equation (2.7) shows that the harvested power scales quadratically with the rotational speed ω , the flux swing $\Delta\Phi$, and the number of coil turns N_c , and linearly with the number of teeth N_t , while it is inversely proportional to the coil resistance R_{Coil} .

For a given application (i.e. fixed ω and N_t), the design of the pickup unit therefore splits into two largely independent optimization problems [13]:

$$P_L \propto \frac{N_c^2}{R_{Coil}}, \quad (2.8a)$$

$$P_L \propto (\Delta\Phi)^2 = \left(\frac{\mathcal{F}_M}{\Delta\mathcal{R}} \right)^2, \quad (2.8b)$$

where $\Delta\mathcal{R} = \mathcal{R}_u - \mathcal{R}_a$ is the difference between the total loop reluctance at the unaligned and aligned positions. Sub-function (2.8a) is a winding-design problem: for a coil of resistivity ρ_{wire} occupying a cross-section of height h_{Coil} and width W_{Coil} inside the pole-piece, it can be shown that (2.8a) reduces to [13]

$$P_L \propto \frac{h_{Coil} W_{Coil} f_w}{\rho_{wire}}, \quad (2.9)$$

with $f_w < 1$ the winding fill factor of the coil; for a fixed available cross-section, output power is thus maximized simply by maximizing the copper fill of the winding window. Sub-function (2.8b), in contrast, is a magnetic-circuit design problem: with the MMF $\mathcal{F}_M = H_c h_M$ fixed by the magnet coercivity H_c and its height h_M , maximizing $\Delta\Phi$ reduces to maximizing the reluctance swing $\Delta\mathcal{R}$ between the two extreme rotor positions.

Magnetic-circuit model of the reluctance swing

The reluctance of a homogeneous flux-path element of length l , cross-section A and relative permeability μ_r is

$$\mathcal{R} = \frac{l}{\mu_0 \mu_r A}. \quad (2.10)$$

Since the soft ferromagnetic pole-piece and toothed wheel have a relative permeability several orders of magnitude larger than that of air ($\mu_r \approx 1$) or of the PM ($\mu_r \approx 1.05$), their contribution to the loop reluctance is negligible, and $\Delta\mathcal{R}$ is dominated by the air gap and by the magnet's own reluctance $\mathcal{R}_M = h_M/(\mu_0\mu_M A_M)$ [13]. Denoting by $\mathcal{R}_{g,a}$ and $\mathcal{R}_{g,u}$ the air-gap reluctance at the aligned and unaligned positions, respectively, the flux at each extreme follows from Hopkinson's law as $\Phi_{max} = \mathcal{F}_M/(\mathcal{R}_{g,a} + \mathcal{R}_M)$ and $\Phi_{min} = \mathcal{F}_M/(\mathcal{R}_{g,u} + \mathcal{R}_M)$, so that the flux swing of (2.8b) can be written explicitly as

$$\Delta\Phi = \mathcal{F}_M \left(\frac{1}{\mathcal{R}_{g,a} + \mathcal{R}_M} - \frac{1}{\mathcal{R}_{g,u} + \mathcal{R}_M} \right). \quad (2.11)$$

Equation (2.11) makes explicit that, once the magnet material and the air-gap length are fixed by mechanical constraints, $\Delta\Phi$ is governed by the pickup-unit geometry surrounding the air gap, namely the tooth height h_{Tooth} (which sets $\mathcal{R}_{g,u}$, since a taller tooth increases the reluctance seen at the unaligned position up to a saturating knee at approximately $h_{Tooth} \approx 3$ mm) and the magnet height h_M (which enters both $\mathcal{F}_M$ and $\mathcal{R}_M$, so that an optimal, non-trivial value of h_M exists) [13].

For a fixed available volume, expressed as a hollow cylinder of overall diameter d_o , shaft diameter d_{Shaft} and depth l_o , the volumetric power density of the harvester is

$$p_v = \frac{P_L}{V_{VREH}}, \quad V_{VREH} = \frac{\pi}{4} l_o (d_o - d_{Shaft})^2, \quad (2.12)$$

and, once the pole pitch and slot width are fixed, the available radial space $C = h_{Coil} + h_{Tooth}$ must be split between the coil and the tooth. Since P_L increases monotonically with h_{Coil} through (2.9) but only up to the knee point through h_{Tooth} in (2.11), for a given C there is an optimal h_{Coil}/h_{Tooth} ratio that maximizes p_v [13]. Similarly, setting $\frac{d\Delta\Phi}{dh_M} = 0$ in (2.11) yields a closed-form optimal magnet height,

$$h_M^* = \mu_M \frac{k_2}{k_1} \frac{W_{g,a}}{W_{g,u}} \sqrt{l_{g,a} l_{g,u}}, \quad (2.13)$$

where $l_{g,a}$, $l_{g,u}$, $W_{g,a}$, $W_{g,u}$ are the effective air-gap length and width at the aligned and unaligned positions, and $k_1, k_2 > 1$ are correction factors accounting for flux leakage and for the MMF drop in the finite-permeability soft iron, respectively [13].

2.1.c Electrical Thevenin Model of the Pickup Unit

The Faraday law derivation of Section 2.1.a gives the ideal electromotive force induced in the coil, but it does not yet describe how that voltage appears at the physical terminals of the pickup unit. A real coil is made of a finite length of copper wire, so it has a nonzero winding resistance, and it carries a current that produces its own magnetic flux, so it also has a nonzero self inductance. Both effects sit in series with the ideal source, since the same wire that generates the electromotive force also carries the resistance and the flux linkage that define R_{Coil} and L_{Coil} .

Circuit equation

Let $e(t)$ denote the ideal electromotive force predicted by (2.3), and let $i(t)$ be the current flowing out of the coil into the external circuit. Applying Kirchhoff's voltage law around the loop formed by the source, the winding resistance, the winding inductance and the external circuit gives

$$e(t) = R_{Coil} i(t) + L_{Coil} \frac{di(t)}{dt} + v(t), \quad (2.14)$$

where $v(t)$ is the voltage measured at the coil terminals, available to the power conditioning circuit. Equation (2.14) is the defining relation of the series generator model adopted throughout this thesis, an ideal voltage source $e(t)$ in series with a resistance R_{Coil} and an inductance L_{Coil} , feeding the external circuit at the coil terminals.

Winding resistance

The winding resistance follows directly from the resistivity of the wire, its total length and its cross sectional area,

$$R_{Coil} = \rho_{wire} \frac{N_c l_t}{A_w}, \quad (2.15)$$

where N_c is the number of turns, l_t is the mean length of a single turn, and A_w is the cross sectional area of the wire. If the coil occupies a winding window of

height h_{Coil} and width W_{Coil} with a copper fill factor f_w , the wire cross section can be written as $A_w = f_w h_{Coil} W_{Coil} / N_c$, so that (2.15) becomes

$$R_{Coil} = \rho_{wire} \frac{N_c^2 l_t}{f_w h_{Coil} W_{Coil}}. \quad (2.16)$$

Treating l_t as approximately constant for a fixed pole piece geometry, (2.16) reproduces the proportionality $N_c^2 / R_{Coil} \propto f_w h_{Coil} W_{Coil} / \rho_{wire}$ already used in (2.9) to optimize the winding cross section [13].

Self inductance and its dependence on rotor position

The self inductance of a coil linked to a magnetic circuit of total reluctance $\mathcal{R}$ is, by the same reluctance formalism used for Hopkinson's law in (2.10),

$$L_{Coil} = \frac{N_c^2}{\mathcal{R}}. \quad (2.17)$$

Because the pickup unit is a variable reluctance device, the reluctance seen by the coil's own flux changes with the rotor angle in exactly the same way as the reluctance that shapes the induced electromotive force. The self inductance therefore swings between two extremes,

$$L_{max} = \frac{N_c^2}{\mathcal{R}_{g,a} + \mathcal{R}_M}, \quad L_{min} = \frac{N_c^2}{\mathcal{R}_{g,u} + \mathcal{R}_M}, \quad (2.18)$$

reached at the aligned and unaligned positions respectively, mirroring the flux extremes already given in (2.11). For circuit level analysis, such as the impedance matching study of Section 2.1.d, this position dependent inductance is normally replaced by a single lumped value L_{Coil} , obtained either as an average over one electrical period or as a direct measurement with an impedance analyzer at the frequency of interest. This is the value referred to as $L_{Coil} \approx 210$ mH later in this chapter.

Phasor representation

For a sinusoidal steady state at the electrical frequency f_c of (2.4), the time domain relation (2.14) can be written in phasor form as

$$E = I (R_{Coil} + j \omega_e L_{Coil}) + V, \quad (2.19)$$

where E , I and V are the phasors associated with $e(t)$, $i(t)$ and $v(t)$, and $\omega_e = 2\pi f_c$. Equation (2.19) shows that, seen from its electrical terminals, the pickup

unit behaves as an ideal source E in series with the impedance

$$Z_s = R_{Coil} + j \omega_e L_{Coil}, \quad (2.20)$$

which is precisely the Thévenin source impedance used in (2.21) for the impedance matching analysis of Section 2.1.d. The series generator, resistance, inductance model derived in this section therefore provides the electrical foundation on which the maximum power transfer condition of the next section, and the power conditioning topologies compared in Chapter ??, are built.

2.1.d Maximum Power Transfer and Impedance Matching

The previous section modelled the pickup unit as a Thévenin source, an open circuit EMF $V_c(t)$ in series with the coil's own impedance. At low electrical frequency this impedance is essentially the winding resistance R_{Coil} , and matching a resistive load to it, $R_L = R_{Coil}$, is enough to extract the maximum available power, as already used in (2.7). As the rotational speed increases, however, the coil inductance L_{Coil} can no longer be neglected, and the condition for maximum power transfer must be generalized. This section derives that general condition and explains why it becomes a central design constraint for a harvester that must operate between 100 and 400 rpm, the target range of this thesis.

2.1.e Thévenin model of the source and the load

Let the coil be represented, at the electrical frequency of interest, by its Thévenin equivalent, an RMS open circuit phasor voltage V_c in series with the source impedance

$$Z_s = R_{Coil} + j \omega_e L_{Coil}, \quad (2.21)$$

where $\omega_e = 2\pi f_c$ is the electrical angular frequency and f_c is the electrical frequency defined in (2.4). The load presented to the coil by the power conditioning circuit is likewise represented, at the same frequency, by a generic complex impedance

$$Z_L = R_L + j X_L. \quad (2.22)$$

Here R_L is the resistive part of the load and X_L its reactive part, which can be inductive ($X_L > 0$) or capacitive ($X_L < 0$) depending on how the load is realized.

2.1.f Power delivered to the load

With Z_s and Z_L in series, the RMS current flowing in the loop is

$$I = \frac{V_c}{Z_s + Z_L}, \quad (2.23)$$

and the average power dissipated in the resistive part of the load is

$$P_L = |I|^2 R_L = \frac{|V_c|^2 R_L}{(R_{Coil} + R_L)^2 + (\omega_e L_{Coil} + X_L)^2}. \quad (2.24)$$

Equation (2.24) is the generalization of (2.7) to a complex load. Setting $L_{Coil} = 0$ and $X_L = 0$ recovers the purely resistive case already used earlier in this chapter.

2.1.g The complex conjugate matching condition

Equation (2.24) can be maximized in two independent steps, since R_L and X_L appear in separate terms of the denominator.

First, for any fixed R_L , the denominator is minimized, and P_L is therefore maximized, by cancelling the reactive term. This requires

$$X_L = -\omega_e L_{Coil}. \quad (2.25)$$

With this choice, (2.24) reduces to

$$P_L = \frac{|V_c|^2 R_L}{(R_{Coil} + R_L)^2}. \quad (2.26)$$

Second, differentiating (2.26) with respect to R_L and setting the result to zero,

$$\frac{\partial P_L}{\partial R_L} = |V_c|^2 \frac{R_{Coil} - R_L}{(R_{Coil} + R_L)^3} = 0 \implies R_L = R_{Coil}. \quad (2.27)$$

Combining (2.25) and (2.27) gives the well known maximum power transfer theorem in its general, complex form,

$$Z_L = Z_s^* = R_{Coil} - j \omega_e L_{Coil}. \quad (2.28)$$

The load must present a resistance equal to the coil resistance and a reactance equal in magnitude but opposite in sign to the coil reactance. Physically, the load must absorb energy from the coil (resistive part) while returning, every half cycle, exactly the reactive energy that the coil inductance would otherwise store (capacitive part). Substituting (2.28) into (2.24) gives the maximum available

power,

$$P_{L,\max} = \frac{|V_c|^2}{4 R_{Coil}}, \quad (2.29)$$

which is independent of L_{Coil} . Once the reactance is properly cancelled, the maximum extractable power depends only on the coil resistance, exactly as in the low frequency case of (2.7).

2.1.h Realizing the reactive compensation

Condition (2.25) shows that an inductive coil requires a load with an equal and opposite *capacitive* reactance. The simplest way to obtain this is a compensation capacitor C_m placed in series with the resistive load, whose reactance at ω_e is $X_{C_m} = -1/(\omega_e C_m)$. Imposing $X_{C_m} = -\omega_e L_{Coil}$ gives the required capacitance

$$C_m = \frac{1}{\omega_e^2 L_{Coil}}. \quad (2.30)$$

Equation (2.30) is the condition used, for instance, to compensate the $L_{Coil} \approx 210$ mH winding discussed at the end of Section 2.1.d. At a given speed, a single series capacitor sized according to (2.30) restores the resistive matching condition of (2.27) and recovers the maximum power of (2.29).

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Wish to say “Thank you”? Write it in *docs/acknowledgements.txt*.

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